FW-EC-PROBE C0 | connection plan / engineering draft

Interfaces agreed; PCB routing, enclosure fit and assembled isolation are not released.

Wetted probe

4 electrodes, K about 1 / cm

Outer electrode F+

Inner electrode S+

Inner electrode S-

Outer electrode F-

Pt1000: four separate wires

- TF+ J3.1 Force high
- TS+ J3.2 Sense high
- TS- J3.3 Sense low
- TF- J3.4 Force low / bias

Sensor lead colours depend on the supplier.

Frontend / electrode domain

ADuCM360 + original Rev A AFE

J5.1 Force+

J5.2 Sense+

J5.3 Sense-

J5.4 Force-

J5.5 = isolated measurement GND
NOT the cable shield / metal shell

Two-wire mode: bridge BOTH pairs
JP401: F+ to S+; JP402: F- to S-
Leave both OPEN for four electrodes.

No Pt100 / 2-wire / 3-wire RTD mode.

Rear power + isolated RS-485

Cable side and electrode side separate

1: 5V isolated (VCC)

2: power return PGND

3: UART TX from MCU

4: UART RX to MCU

5: RS485 DE

6: RS485 /RE

7: 3.3V filtered

8: quiet return GND

J402 (frontend) <-> J202 (rear), pin-for-pin
Pins 5 + 6 share MCU P0.5. Do not short 2 + 8 here.

Minimum SWD pads: J401

1 = VTref 3.3 V; 2 = SWDIO; 3 = SWCLK; 4 = /RESET; 5 = GND

VTref is a debugger reference, not an external power input.

An earthed debugger can bypass electrode-domain isolation during bench work.

Burn / calibrate through the electrode side; verify the complete isolation setup.

Only conductivity, temperature and Modbus RTU. No LCD, keys, buzzer or LEDs.

External cable: J201

- 1 RED: +12 to +30 V DC
- 2 YELLOW: supply return
- 3 BLUE: RS-485 A
- 4 GREEN: RS-485 B
- 5 SHIELD: metal housing only

Default 9600 / 8N1. 120 ohm termination is optional.

Do not connect cable return / shield to electrode GND or PGND.